Mini-project — High-temperature heat pump cycles

Design and comparison of advanced cycles

1. Introduction

A **heat pump** upgrades heat from a low-temperature source to a higher-temperature sink, consuming compression work. Its efficiency is the **coefficient of performance** (COP): the useful heat delivered divided by the work supplied.

A **high-temperature heat pump (HTHP)** delivers heat at a high sink temperature (here around 130 °C) from a modest source. This large temperature lift is demanding: it raises the compression ratio, pushes the **compressor discharge temperature** up (a hard limit for lubricants and materials), and lowers the COP. The standard single-stage cycle quickly reaches its limits.

Advanced cycles address these limits — by splitting the compression, recovering internal heat, injecting vapour, or cascading two circuits — to raise the COP and/or to lower the discharge temperature. The discharge temperature is not a free input: it is **set by the cycle and its parameters**; the groups do not choose it but read it as a result and compare it across variants (see §3.4). This project asks each group to quantify what one such cycle brings compared with the standard one.

2. Context

Each group studies **two systems**: a common **baseline cycle S0** (studied by everyone) and **one advanced cycle** (one per group). The goal is a clear **base-vs-advanced comparison**; at the end, the groups pool their results and compare the advanced cycles against the shared baseline.

All systems share the same case study and the same fixed parameters (Section 3), so the comparison is meaningful. The work is in **design mode only**: all operating conditions are fixed, and the cycle is solved at a single steady-state operating point.

3. Common specification and assumptions

Case study

High-temperature heat pump recovering waste heat to a hot water sink.

- **Sink (condenser)**: water heated from 120 °C to 130 °C, at $p_{\text{sink}} = 10$ bar (kept liquid).
- **Source (evaporator)**: waste-heat water cooled from 35 °C to 30 °C, at $p_{\text{source}} = 3$ bar.
- **Heating capacity**: $Q = 100$ kW delivered at the condenser (for S2, the high-temperature-circuit condenser to the water); this fixes the refrigerant mass flow(s).

Fixed parameters

Parameter	Value	Note
Refrigerant (S0, S1, S3, S4, S5)	R1233zd(E)	low-GWP HT fluid; fallback R245fa / R600
Refrigerant (S2 cascade)	R290 / R1233zd(E)	low-temperature / high-temperature circuit
Compressor isentropic efficiency	0.70	same for every stage
Superheat at evaporator outlet	5 K	S4 excepted (see S4)
Subcooling at condenser outlet	5 K	see S2 exception
Pinch (evaporator, condenser)	5 K	on the water-outlet terminal
Cascade HEX pinch (S2)	5 K	
Ambient temperature (S1 intercooler)	15 °C	air-cooled
Film coefficient — refrigerant	3000 W/m^{2}K	condensing / evaporating / liquid
Film coefficient — water	3000 W/m^{2}K	secondary water circuits
Film coefficient — air	60 W/m^{2}K	S1 intercooler, gas side
Area ratio (cold : hot)	1 : 1	except air-cooled intercooler: 15 : 1 (finned air side)
Fluid properties	CoolProp	

The film coefficients are common to all groups: they turn each exchanger duty into an area, so that the **heat-exchanger sizes** are comparable between variants.

Saturation temperatures. The terminal pinch is taken on the water-outlet side, which fixes the two saturation pressures:

$$T_{\text{cond}} = T^{\text{sink}}_{\text{water,out}} + 5\ \text{K} = 135\ ^\circ\text{C} \ \Rightarrow\ P_{\text{cond}}, \qquad T_{\text{evap}} = T^{\text{source}}_{\text{water,out}} - 5\ \text{K} = 25\ ^\circ\text{C} \ \Rightarrow\ P_{\text{evap}}.$$

Modelling assumptions

A. Design mode only — no simulation mode (all conditions fixed). Steady-state resolution (`solve`).

B. Heat exchangers: single-zone `SizingHX` with constant c_p, used as-is including for the two-phase condenser and evaporator. This is an assumed approximation (constant c_p across the phase change) and must be stated explicitly in the code/comments.

C. Expansion: an isenthalpic `Valve` component is used; its outlet pressure is imposed by a system-level constraint (`set_constraint`). The valve flow law does not affect the fixed design point (the cascade S2 happens to use the `linear` law).

D. New components to create: `FlashTank` (two-phase separator at P_{int}: saturated liquid + saturated vapour), `Mixer` (injected vapour + low-pressure discharge at P_{int}), and `Splitter` (one liquid inlet → two outlets at the same state, used by S5).

E. IHX (S4) and economizer (S5): approximate single-zone models (the multi-zone model is no longer functional).

F. Intermediate pressure (S1, S3, S5): default $P_{\text{int}} = \sqrt{P_{\text{evap}}\,P_{\text{cond}}}$ (geometric mean of pressures). This does not apply to the cascade (S2), whose intermediate *temperature* rule is given in S2.

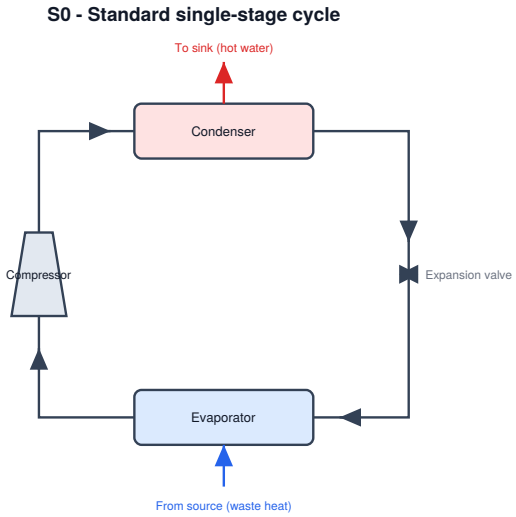
Comparison metrics

At the fixed design point, each group reports and compares against S0:

- **Heating COP** $= Q_{\text{cond}}/W_{\text{comp}}$ (sum of all compressors);

- **Compressor discharge temperature** — a **computed result** (set by the cycle, not an input): reported and compared across variants as a high-temperature operating-limit indicator, on the same footing as the COP;
- **Heat-exchanger sizes** (areas, from the design solve) — compared on a common basis via the film coefficients above;
- **Compression ratio** per stage.

4. The baseline cycle — S0



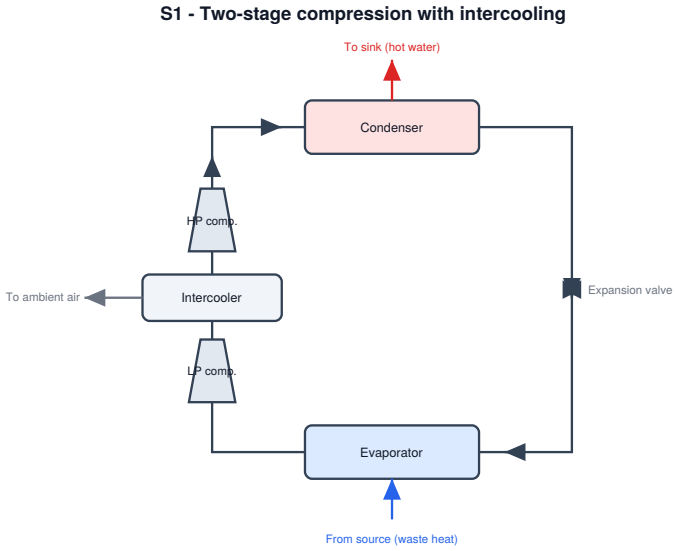
The standard single-stage vapour-compression cycle, with four components: **compressor**, **condenser**, **expansion valve**, **evaporator**. The refrigerant evaporates at T_{evap} (with 5 K superheat at outlet), is compressed to P_{cond}, condenses at T_{cond} (with 5 K subcooling at outlet), and is expanded back to P_{evap}.

S0 is built by every group and serves as the reference against which the advanced cycle is compared.

5. Advanced cycle specifications

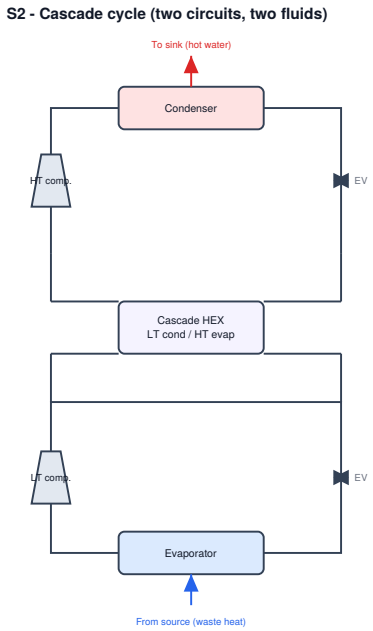
Each group develops one of the following, in addition to S0. The common specification (Section 3) applies throughout; only the points below are specific.

S1 — Two-stage compression with intercooling



Two compressors with an **intercooler** rejecting heat to ambient air between the stages, at the intermediate pressure $P_{\text{int}} = \sqrt{P_{\text{evap}}\,P_{\text{cond}}}$. The intercooler is a real air heat exchanger with $T_{\text{amb}} = 15\,^{\circ}\text{C}$; the air leaves at $25\,^{\circ}\text{C}$ (fixed). The refrigerant intercooler-outlet temperature is a **parameter to optimise**; by default it is $T_{\text{sat}}(P_{\text{int}}) + 5$ K (superheated vapour, consistent with the evaporator superheat), within the 5 K pinch limit of the exchanger. Expected effect: lower compression work and a lower final discharge temperature.

S2 — Cascade cycle

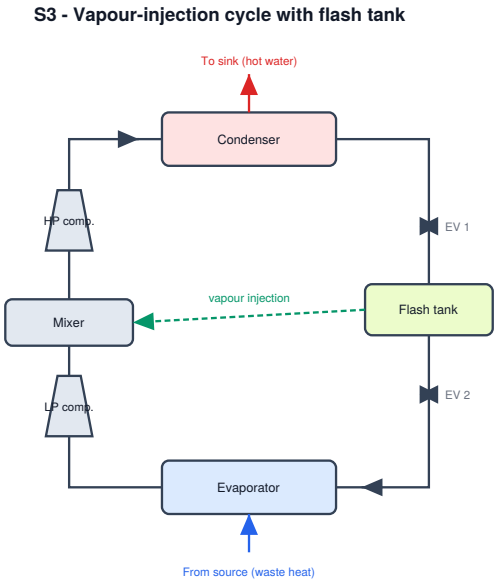


Two circuits coupled by a **cascade heat exchanger**: a low-temperature (LT) circuit with R290 and a high-temperature (HT) circuit with R1233zd(E), each with its own compressor and expansion valve. The intermediate temperature T_{mid} is a **free parameter to optimise** during the

project; its default value is the geometric mean $T_{\text{mid}} = \sqrt{T_{\text{evap}}\, T_{\text{cond}}}$ (in kelvin, $\approx 76\,^{\circ}\text{C}$ with the nominal levels). The LT circuit condenses at T_{mid}; the HT circuit evaporates at $T_{\text{mid}} - 5$ K (cascade pinch).

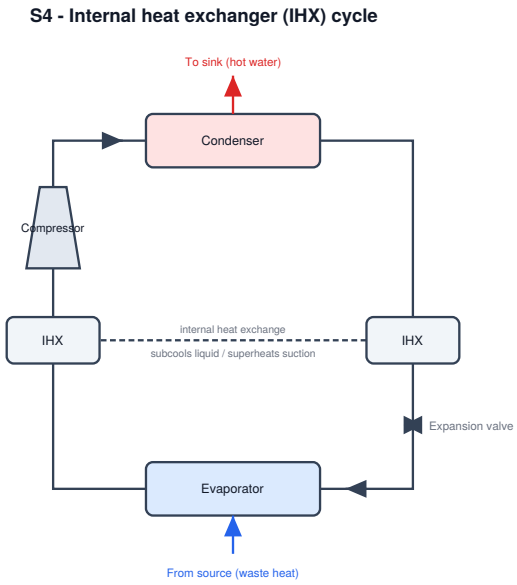
Important exception on subcooling. No subcooling is applied to the LT circuit at the cascade HEX: the LT condenser delivers **saturated liquid** at T_{mid}, instead of the 5 K subcooling applied elsewhere. The reason is an area constraint: with the 5 K cascade pinch between $T_{\text{cond,LT}}$ and $T_{\text{evap,HT}}$, adding 5 K of LT subcooling would bring the LT liquid outlet down to exactly the HT evaporation temperature, giving $\Delta T = 0$ at the cold end of the cascade HEX — an infinite LMTD / infinite area, physically infeasible. Delivering saturated liquid keeps a real 5 K ΔT at both ends. The 5 K subcooling is still applied at the HT condenser (sink side). Note the (small) COP penalty this entails, since the LT liquid enters the LT evaporator "warmer".

S3 — Vapour-injection cycle with flash tank



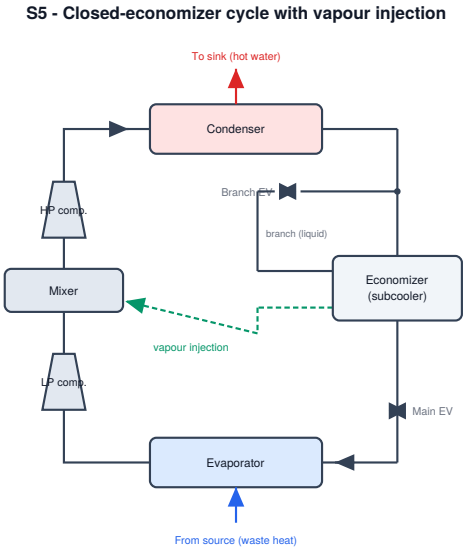
Two expansion valves with a **flash tank** at P_{int} between them. The two-phase fluid leaving valve 1 is separated: saturated liquid goes to valve 2 and then the evaporator; saturated vapour is injected into the compressor. As there is no dedicated injection-compressor model, use **two compressor models**: a low-pressure stage ($P_{\text{evap}} \to P_{\text{int}}$), a `Mixer` that blends the injected vapour with the LP discharge at P_{int}, then a high-pressure stage ($P_{\text{int}} \to P_{\text{cond}}$). Mind the **mass balance**: the injected vapour fraction depends on the flash quality, so the LP and HP mass flows differ.

S4 — Internal heat exchanger (IHX)



An **IHX** between the liquid line (condenser outlet) and the suction line (evaporator outlet): it subcools the high-pressure liquid and superheats the low-pressure suction. Model it as an approximate single-zone exchanger. Because the IHX itself provides the suction superheat, the two cycle closures move: the **evaporator outlet** is imposed at a **vapour quality** x_{evap} (default 0.90, i.e. two-phase — it replaces the usual 5 K evaporator superheat), and the **suction superheat** ΔT_{suc} (default 5 K) is imposed at the **IHX cold outlet** (compressor inlet). Both x_{evap} and ΔT_{suc} are the knobs of this cycle. Expected effect: increased refrigerating effect at the cost of a higher suction temperature (hence discharge temperature) — a trade-off to quantify. Note: with $x_{\text{evap}} = 1$ and the default suction superheat, S4 reduces to S0; the point of S4 is precisely the **parametric sweep** of x_{evap} and ΔT_{suc}, not the single default point.

S5 — Closed economizer



A **subcooler heat exchanger** (closed economizer) instead of a flash tank, with vapour injection into the compressor. A fraction of the liquid is expanded to P_{int} and evaporated in the economizer,

subcooling the main liquid stream; that vapour is injected at P_{int}. Model with two compressor stages and a `Mixer` as in S3, plus a `Splitter` to bleed the injection fraction; model the economizer as an approximate single-zone exchanger. Economizer closures: a 5 K pinch on the liquid side (the main-liquid outlet leaves at $T_{\text{sat}}(P_{\text{int}}) + 5$ K), and the injected stream leaves the economizer as **saturated vapour** at P_{int}. Compare flash-tank injection (S3) and closed-economizer injection (S5).

6. Expected deliverables

This is an **open project**: beyond the fixed design point, each group is free to propose analyses, parameter changes or circuit modifications that improve performance. Each group submits **two deliverables**:

1. **A project report** (max. 10 pages) presenting the model and an analysis of the results: COP, compressor discharge temperature, heat-exchanger sizes, compression ratios, and energy/exergy balances — with a clear **base-vs-advanced comparison**.
2. **Python code**: commented, in a **single file** (notebook or `.py`) containing the model and the analyses presented in the report. The model is built on the shared library and reuses the components listed in Section 3 (`FlashTank`, `Mixer`, `Splitter`) where required.

Examples of extensions useful for the open-ended work include: parametric studies of the intermediate pressure P_{int}, of the superheat / subcooling, of the pinch values, of the intercooler refrigerant-outlet temperature (S1), of the IHX knobs x_{evap} / ΔT_{suc} (S4), or of the cascade intermediate temperature T_{mid} (S2); adding or changing components (e.g. combining an IHX with another cycle, a second injection, additional subcooling); alternative refrigerants within the high-temperature range; or any other modification, justified physically and quantified with the model. The quality of the reasoning and of the quantified comparison matters more than finding a single "best" answer.